

The Determinant

Proposition :

There is at most one
function

$$\det : M_{n \times n}(F) \longrightarrow F$$

with the following properties:

(Identity) (1) $\det(I_n) = 1$

(2) $\det \begin{pmatrix} v_1^T \\ \vdots \\ v_i^T \\ v_i^T \\ v_i^T \\ \vdots \\ v_n^T \end{pmatrix} = 0$

(Alternating)

$\leftarrow i\text{-th row}$
 $\leftarrow (i+1)\text{-st row}$

(2')

$$\det \begin{pmatrix} v_1^T \\ \vdots \\ v_i^T \\ v_{i+1}^T \\ \vdots \\ v_n^T \end{pmatrix} = - \det \begin{pmatrix} v_1^T \\ \vdots \\ v_{i+1}^T \\ v_i^T \\ \vdots \\ v_n^T \end{pmatrix}$$

(3)

$$(c) \det \begin{pmatrix} v_1^T \\ \vdots \\ k \cdot v_i^T \\ \vdots \\ v_n^T \end{pmatrix} = k \cdot \det \begin{pmatrix} v_1^T \\ \vdots \\ v_i^T \\ \vdots \\ v_n^T \end{pmatrix}$$

(Multi-linearity)

$k \in F$

(Scaling)

(r)

(Addition)

$$\det \begin{pmatrix} v_1^T \\ \vdots \\ v_i^T + w_i^T \\ \vdots \\ v_n^T \end{pmatrix}$$

$$= \det \begin{pmatrix} v_1^T \\ \vdots \\ v_i^T \\ \vdots \\ v_n^T \end{pmatrix} + \det \begin{pmatrix} v_1^T \\ \vdots \\ w_i^T \\ \vdots \\ v_n^T \end{pmatrix}$$

Rmd:

$$(2) + 3(r) \Rightarrow (2')$$

Example:

$$\det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix} \stackrel{?}{=} - \det \begin{pmatrix} v_2^T \\ v_1^T \end{pmatrix}$$

$$\begin{aligned} \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix} &= \det \begin{pmatrix} v_1^T + v_2^T \\ v_2^T \end{pmatrix} \\ &\stackrel{||}{=} \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix} + \det \begin{pmatrix} v_2^T \\ v_2^T \end{pmatrix} \end{aligned}$$

$$\det \begin{pmatrix} v_1^T + v_2^T \\ v_1^T + v_2^T \end{pmatrix} = \det \begin{pmatrix} v_1^T \\ v_1^T + v_2^T \end{pmatrix} + \det \begin{pmatrix} v_2^T \\ v_1^T + v_2^T \end{pmatrix}$$

$$= \det \begin{pmatrix} v_1^T \\ v_1^T \end{pmatrix} + \det \begin{pmatrix} v_1^T \\ v_2^T \end{pmatrix}$$

$$+ \det \begin{pmatrix} \begin{bmatrix} v_2^T \\ \vdots \\ v_i^T \end{bmatrix} \end{pmatrix} + \det \begin{pmatrix} \begin{bmatrix} v_2^T \\ \vdots \\ v_i^T \end{bmatrix} \end{pmatrix}$$

Key point for showing the uniqueness of det:

det has specified behavior with respect to row operations:

(1) (Swapping)

$$\det \begin{pmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_i^T \\ \vdots \\ v_j^T \\ \vdots \\ v_n^T \end{bmatrix} \end{pmatrix} = = \det \begin{pmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_j^T \\ \vdots \\ v_i^T \\ \vdots \\ v_n^T \end{bmatrix} \end{pmatrix}$$

(2) Adding a multiple of one row to another

$$\det \begin{pmatrix} v_1^T \\ \vdots \\ v_i^T + k \cdot v_j^T \\ \vdots \\ v_j^T \\ \vdots \\ v_n^T \end{pmatrix} = \det \begin{pmatrix} v_1^T \\ \vdots \\ v_i^T \\ \vdots \\ v_j^T \\ \vdots \\ v_n^T \end{pmatrix}$$

[Also the same behavior if $i > j$]

(3) Scaling

Same as 3(a) above

These properties pin down $\det$ completely.

A

$\left\{ \begin{array}{l} \text{row} \\ \text{operations} \end{array} \right\}$ we know how
det changes along
these operations
 $\text{RREF}(A)$

So $\det(\text{RREF}(A))$ determines in an
algorithmic way $\det(A)$.

There are two possibilities:

(a) $\text{RREF}(A) = I_n \Leftrightarrow A$ is invertible

In this case

$$\det(\text{RREF}(A)) = 1$$

$$\Rightarrow \det(A) \neq 0$$

(b) $\text{RREF}(A) \neq I_n \Leftrightarrow A$ is non-invertible

In this case

$[I, I]$

$$\text{REF}(A) = \begin{bmatrix} v_1^T \\ \vdots \\ v_j^T \\ \vdots \\ 0^T \end{bmatrix}$$

$$\Rightarrow \det(\text{REF}(A)) = 0 \cdot \det(\text{REF}(A)) = 0$$

$$\Rightarrow \det(A) = 0$$

Example:

$$\det \begin{pmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 1 \\ 2 & 1 & 3 \end{bmatrix} \end{pmatrix}$$

no change $\downarrow R_3 - 2 \cdot R_1$

$$\det \begin{pmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 1 \\ 0 & -3 & -3 \end{bmatrix} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -3 \\ 0 & 1 & 1 \end{pmatrix}$$

$-3 \rightarrow -\frac{1}{3} \cdot \det \downarrow R3 \times -\frac{1}{3}$

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$-1 \rightarrow -1 \cdot \det \downarrow R2 \leftrightarrow R3$

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 5 & 1 \end{pmatrix}$$

$\text{no change} \downarrow R1 - 2 \times R2$
 $\downarrow R3 - 5 \times R2$

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -4 \end{pmatrix}$$

$$-4 \left[-\frac{1}{4} \cdot \det \downarrow -\frac{1}{4} \times R_3 \right]$$

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$



no
change

$$1 = \det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\Rightarrow$ det of original matrix

$$= -12$$

Proposition (2) $A, B \in M_{n \times n}(F)$

$$\det(A B) = \det(A) \det(B)$$

Idea: Both sides of the equation

satisfy properties (2) & (3) (c) & (3) (d)

of the determinant function for

fixed B & varying A

Proposition (3)

$$\det(A^T) = \det(A)$$